

Chapter 5 – Initial Server Configuration

Objective

After installing Ubuntu Server, several configuration tasks should be completed before deploying additional software.

In this chapter, we will:

- Update Ubuntu
- Upgrade installed packages
- Verify the operating system version
- Check network connectivity
- Verify system resources
- Install essential packages
- Prepare the server for future software such as Wazuh

5.1 Login to Ubuntu Server

After the installation completes, log in using the username and password created during setup.

Example:

Ubuntu 22.04.5 LTS

wazuh-server login:

Username:

kashifkhan

Password:

After logging in successfully, you should see:

kashifkhan@wazuh-server:~\$

This confirms the server is ready for administration.

5.2 Update Package Information

Before installing any software, update the package index.

Command:

```
sudo apt update
```

Understanding the Command

sudo

Runs the command with administrator privileges.

apt

Ubuntu's package manager.

update

Downloads the latest package information from Ubuntu repositories.

It **does not install updates**; it only refreshes the package list.

Why Update First?

Updating the package list ensures Ubuntu knows about the latest software versions and security patches available.

Running apt update before installing software is considered a Linux best practice.

```

<14>Jul 16 05:31:42 cloud-init: 256 SHA256:ccy9SHGPsFQFGx63U6gncAr3Nyx0ektnm/G104IemjM root@ubunv22
(ECDSA)
<14>Jul 16 05:31:42 cloud-init: 256 SHA256:HX7H6GvLIpr05KPQKFZzq03bpWKhhR0XJ0Ha6/qQKr8 root@ubunv22
(ED25519)
<14>Jul 16 05:31:42 cloud-init: 3072 SHA256:iqV0XPiknfK5As9ER4DYt1+8P5/wTjoq54bjse2cq7g root@ubunv22
(RSA)
<14>Jul 16 05:31:42 cloud-init: -----END SSH HOST KEY FINGERPRINTS-----
<14>Jul 16 05:31:42 cloud-init: #####
-----BEGIN SSH HOST KEY KEYS-----
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBKIVqhVYVgJAcPWUQXpnqneuQ0jh
N17qvq7EWvf6g5bTUIPku1dTE1rgLoaiGNinQ0H0PkAg7sIQbuKiPCmRzVc= root@ubunv22
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIK7LtkSMRTUkmRpC5DqWEJ9248XyZfxgP6EFb+3UVH2+ root@ubunv22
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQC8CzS10I2b/KST+r/Ufb01v4Fkw0+W3hEc4IOd9UjWejXz12C02zsLdpQhGaxx
sJjvxLHFgZ0R3Vz3GM6vCCKVVAx0u5+gET6Ew/PY7YeB00w6Gxlygz42efCGHJsbpuhCmJ0DiRBdw2Ef9gYH1u4ShFaJf8V/gB00
9PbXNxcgC1EsLpUDUwKdjEnX6eKy2GUBjE43sXM3r2wjXXrfgp67f7HEt6e/v1HFodi+p1p1B+uC/VknNKmSuygUNKB8JyRZYkDU
LJNWpkKnyIMkPS0kzIiuZp9DaMao17RFEH7wPdwV30pId1e2tZ6PSLCwdptYh1M4e2uY1CK0+pQRF652NzqyyVwXQCWdm8Rh/Qr
HB9mpUYcXWNUqJ6UTZrzVLKkh6yRYKPBvoPK0SG+SkDfApURm/pz2UQ9FHRI88TS9oWqX+Qv8VzaMMWUL1qBG9G0jBNWF i/qIgvDU
EiD/oTilIakKnDzofkox2wBA0/5p7afYMrvXJMjZCRvVjNtPEck= root@ubunv22
-----END SSH HOST KEY KEYS-----
[ 77.916022] cloud-init[1713]: Cloud-init v. 25.1.4-0ubuntu0~22.04.1 finished at Thu, 16 Jul 2026
05:31:42 +0000. Datasource DataSourceNone. Up 77.90 seconds
[ OK ] Finished Cloud-init: Final Stage.
[ OK ] Reached target Cloud-init target.

kashif@ubunv22:~$ sudo apt update
[sudo] password for kashif:
Sorry, try again.
[sudo] password for kashif:
Hit:1 http://pk.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://pk.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://pk.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
63 packages can be upgraded. Run 'apt list --upgradable' to see them.
kashif@ubunv22:~$

```

5.3 Upgrade Installed Packages

Once the package list is updated, install available updates.

Command:

```
sudo apt upgrade -y
```

Understanding the Command

upgrade

Downloads and installs updated versions of installed packages.

-y

Automatically answers **Yes** to installation prompts.

```

Preparing to unpack .../08-ubuntu-drivers-common_1%3a0.9.6.2~0.22.04.11_amd64.deb ...
Unpacking ubuntu-drivers-common (1:0.9.6.2~0.22.04.11) over (1:0.9.6.2~0.22.04.8) ...
Preparing to unpack .../09-iproute2_5.15.0-1ubuntu2.2_amd64.deb ...
Unpacking iproute2 (5.15.0-1ubuntu2.2) over (5.15.0-1ubuntu2) ...
Preparing to unpack .../10-libapparmor1_3.0.4-2ubuntu2.5_amd64.deb ...
Unpacking libapparmor1:amd64 (3.0.4-2ubuntu2.5) over (3.0.4-2ubuntu2.4) ...
Preparing to unpack .../11-libcryptsetup12_2%3a2.4.3-1ubuntu1.3_amd64.deb ...
Unpacking libcryptsetup12:amd64 (2:2.4.3-1ubuntu1.3) over (2:2.4.3-1ubuntu1.2) ...
Preparing to unpack .../12-initramfs-tools_0.140ubuntu13.5_all.deb ...
Unpacking initramfs-tools (0.140ubuntu13.5) over (0.140ubuntu13.4) ...
Preparing to unpack .../13-initramfs-tools-core_0.140ubuntu13.5_all.deb ...
Unpacking initramfs-tools-core (0.140ubuntu13.5) over (0.140ubuntu13.4) ...
Preparing to unpack .../14-initramfs-tools-bin_0.140ubuntu13.5_amd64.deb ...
Unpacking initramfs-tools-bin (0.140ubuntu13.5) over (0.140ubuntu13.4) ...
Preparing to unpack .../15-linux-base_4.5ubuntu9+22.04.1_all.deb ...
Unpacking linux-base (4.5ubuntu9+22.04.1) over (4.5ubuntu9) ...
Preparing to unpack .../16-netplan.io_0.107.1-3ubuntu0.22.04.4_amd64.deb ...
Unpacking netplan.io (0.107.1-3ubuntu0.22.04.4) over (0.106.1-7ubuntu0.22.04.4) ...
dpkg: warning: unable to delete old directory '/etc/netplan': Directory not empty
Selecting previously unselected package netplan-generator.
Preparing to unpack .../17-netplan-generator_0.107.1-3ubuntu0.22.04.4_amd64.deb ...
Unpacking netplan-generator (0.107.1-3ubuntu0.22.04.4) ...
Selecting previously unselected package python3-netplan.
Preparing to unpack .../18-python3-netplan_0.107.1-3ubuntu0.22.04.4_amd64.deb ...
Unpacking python3-netplan (0.107.1-3ubuntu0.22.04.4) ...
Preparing to unpack .../19-libnetplan0_0.107.1-3ubuntu0.22.04.4_amd64.deb ...
Unpacking libnetplan0:amd64 (0.107.1-3ubuntu0.22.04.4) over (0.106.1-7ubuntu0.22.04.4) ...
Preparing to unpack .../20-ubuntu-pro-client-l10n_37.2ubuntu~22.04_amd64.deb ...
Unpacking ubuntu-pro-client-l10n (37.2ubuntu~22.04) over (33.2~22.04) ...
Preparing to unpack .../21-ubuntu-pro-client_37.2ubuntu~22.04_amd64.deb ...
Unpacking ubuntu-pro-client (37.2ubuntu~22.04) over (33.2~22.04) ...
Preparing to unpack .../22-ubuntu-advantage-tools_37.2ubuntu~22.04_all.deb ...
Unpacking ubuntu-advantage-tools (37.2ubuntu~22.04) over (33.2~22.04) ...
Preparing to unpack .../23-ubuntu-minimal_1.481.5_amd64.deb ...
Unpacking ubuntu-minimal (1.481.5) over (1.481.3) ...

Progress: [ 23%] [#####.....]

```

Difference Between Update and Upgrade

Command	Purpose
<code>sudo apt update</code>	Refreshes the package list.
<code>sudo apt upgrade</code>	Installs available package updates.

Many beginners confuse these two commands.

5.4 Verify Ubuntu Version

Check the installed operating system version.

Command:

`lsb_release -a`

Example output:

Distributor ID: Ubuntu

Description: Ubuntu 22.04.5 LTS

Release: 22.04

Codename: jammy

This confirms the server is running the expected Ubuntu release.

```
Last login: Thu Jul 16 05:49:29 UTC 2026 on tty1
kashif@ubunv22:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:49:9d:94 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86369sec preferred_lft 86369sec
    inet6 fd17:625c:f037:2:a00:27ff:fe49:9d94/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86372sec preferred_lft 14372sec
    inet6 fe80::a00:27ff:fe49:9d94/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:30:bd:69 brd ff:ff:ff:ff:ff:ff
kashif@ubunv22:~$ ping google.com
PING google.com (142.250.187.78) 56(84) bytes of data.
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=1 ttl=255 time=114 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=2 ttl=255 time=189 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=3 ttl=255 time=141 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2013ms
rtt min/avg/max/mdev = 113.681/147.795/188.731/31.016 ms
kashif@ubunv22:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.5 LTS
Release:        22.04
Codename:       jammy
kashif@ubunv22:~$ _
```

5.5 Check Internet Connectivity

Verify that the server can communicate with the Internet.

Command:

ping google.com

Example output:

64 bytes from ...

icmp_seq=1

ttl=117

time=24 ms

Stop the ping with:

Ctrl + C

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jul 16 05:49:29 UTC 2026 on tty1
kashif@ubunv22:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:49:9d:94 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86369sec preferred_lft 86369sec
    inet6 fd17:625c:f037:2:a00:27ff:fe49:9d94/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86372sec preferred_lft 14372sec
    inet6 fe80::a00:27ff:fe49:9d94/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:30:bd:69 brd ff:ff:ff:ff:ff:ff
kashif@ubunv22:~$ ping google.com
PING google.com (142.250.187.78) 56(84) bytes of data.
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=1 ttl=255 time=114 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=2 ttl=255 time=189 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=3 ttl=255 time=141 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2013ms
rtt min/avg/max/mdev = 113.681/147.795/188.731/31.016 ms
kashif@ubunv22:~$ _
```

Why Is This Important?

Internet connectivity is required for:

- Installing software
- Downloading updates

- Accessing repositories
 - Installing Wazuh
 - Installing Docker
-

5.6 Verify Network Configuration

Display all network interfaces.

Command:

```
ip addr
```

This command displays:

- Network interfaces
- IP addresses
- Interface status

Example:

```
enp0s3
```

```
10.0.2.15
```

At this stage, DHCP is acceptable.

Static IP configuration will be covered later.

```

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jul 16 05:49:29 UTC 2026 on tty1
kashif@ubunv22:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:49:9d:94 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86369sec preferred_lft 86369sec
    inet6 fd17:625c:f037:2:a00:27ff:fe49:9d94/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86372sec preferred_lft 14372sec
    inet6 fe80::a00:27ff:fe49:9d94/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:30:bd:69 brd ff:ff:ff:ff:ff:ff
kashif@ubunv22:~$ ping google.com
PING google.com (142.250.187.78) 56(84) bytes of data:
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=1 ttl=255 time=114 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=2 ttl=255 time=189 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=3 ttl=255 time=141 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2013ms
rtt min/avg/max/mdev = 113.681/147.795/188.731/31.016 ms
kashif@ubunv22:~$ _

```

5.7 Check Disk Usage

Command:

`df -h`

Purpose:

Displays:

- Total storage
- Used storage
- Available storage
- Mounted file systems

Always verify available disk space before installing server applications.

5.8 Check Memory Usage

Command:

`free -h`

Displays:

- Total RAM
- Used RAM
- Free RAM
- Swap

This helps determine whether the server meets the memory requirements for additional software.

```
link/ether 08:00:27:49:9d:94 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
    valid_lft 86369sec preferred_lft 86369sec
inet6 fd17:625c:f037:2:a00:27ff:fe49:9d94/64 scope global dynamic mngtmpaddr noprefixroute
    valid_lft 86372sec preferred_lft 14372sec
inet6 fe80::a00:27ff:fe49:9d94/64 scope link
    valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:30:bd:69 brd ff:ff:ff:ff:ff:ff
kashif@ubunv22:~$ ping google.com
PING google.com (142.250.187.78) 56(84) bytes of data.
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=1 ttl=255 time=114 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=2 ttl=255 time=189 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=3 ttl=255 time=141 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2013ms
rtt min/avg/max/mdev = 113.681/147.795/188.731/31.016 ms
kashif@ubunv22:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.5 LTS
Release:        22.04
Codename:       jammy
kashif@ubunv22:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            494M  1.1M  493M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv  12G   3.3G   7.4G  31% /
tmpfs            2.5G   0  2.5G   0% /dev/shm
tmpfs            5.0M   0  5.0M   0% /run/lock
/dev/sda2        2.0G  133M   1.7G   8% /boot
tmpfs            494M  4.0K  494M   1% /run/user/1000
kashif@ubunv22:~$ free -h
              total        used        free       shared  buff/cache       available
Mem:           4.8Gi       175Mi       4.4Gi         1.0Mi        230Mi        4.4Gi
Swap:            0B           0B           0B
```

5.9 Install Essential Packages

Install common administration tools.

Command:

```
sudo apt install -y \
```

```
curl \
```

```
wget \
```

```
vim \
```

```
nano \
```

```
git \
```

```
net-tools
```

Package Overview

Package	Purpose
curl	Download files from web servers
wget	Download files via HTTP/HTTPS
vim	Advanced text editor
nano	Beginner-friendly text editor
git	Version control system
net-tools	Provides commands such as ifconfig

These tools are useful for nearly every Linux server.

5.10 Verify Hostname

Command:

hostnamectl

Displays:

- Hostname
- Operating System
- Kernel Version
- Architecture

Example:

Hostname:

Ubunv22

A meaningful hostname makes server identification easier.

```
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2013ms
rtt min/avg/max/mdev = 113.681/147.795/188.731/31.016 ms
kashif@ubunv22:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:   Ubuntu 22.04.5 LTS
Release:      22.04
Codename:     jammy
kashif@ubunv22:~$ df -h
Filesystem                Size      Used Avail Use% Mounted on
tmpfs                      494M        1.1M  493M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 12G        3.3G    7.4G  31% /
tmpfs                      2.5G         0    2.5G   0% /dev/shm
tmpfs                      5.0M         0    5.0M   0% /run/lock
/dev/sda2                  2.0G       133M    1.7G   8% /boot
tmpfs                      494M        4.0K   494M   1% /run/user/1000
kashif@ubunv22:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           4.8Gi        175Mi        4.4Gi         1.0Mi         230Mi        4.4Gi
Swap:           0B           0B           0B
kashif@ubunv22:~$ hostsystemctl
hostsystemctl: command not found
kashif@ubunv22:~$ hostnamectl
  Static hostname: ubunv22
        Icon name: computer-vm
        Chassis: vm
        Machine ID: 25ae4f3ccee74aae8b008ae05e20ad26
        Boot ID: 58223a17da244df88e511d3ef54eea6b
  Virtualization: oracle
Operating System: Ubuntu 22.04.5 LTS
        Kernel: Linux 5.15.0-185-generic
  Architecture: x86_64
  Hardware Vendor: innotek GmbH
  Hardware Model: VirtualBox
kashif@ubunv22:~$ _
```

Best Practices

- Update Ubuntu immediately after installation.
 - Install security updates regularly.
 - Verify internet connectivity.
 - Record your hostname and IP address.
 - Install only necessary packages.
-

Chapter Summary

In this chapter, you learned how to:

- Update package information.
 - Upgrade installed software.
 - Verify the Ubuntu version.
 - Test Internet connectivity.
 - Check networking.
 - Monitor disk and memory usage.
 - Install essential administration tools.
 - Verify the server hostname.
-

One-Line Summary

A newly installed Ubuntu Server should always be updated, verified, and equipped with essential administration tools before additional services or applications are deployed.
